CST1500 Coursework Two – Python Group Project

Deadline for submission: Friday 13th December 2024 18:00 (Week 12).

NOTE: Due to expected high use, there will be a 2-hour window where submissions will be accepted and will not incur a late penalty. Submissions received after 20:00 will be treated as late and will lose 10% of their final mark (to a limit of no lower than 40%, providing the final mark before penalties is above 40%). Submissions received more than 24 hours late will receive zero marks. DO NOT TREAT THIS AS A 20:00 DEADLINE – I advise starting your submission as early as possible before the 18:00 deadline.

Please read all the assignment specification carefully first, before asking your tutor any questions.

General Information

The second piece of coursework for CST1500 is a Group Project. Your group should design and implement a piece of software using the Python programming language.

IMPORTANT

- This is a **GROUP** Coursework and will be completed in groups of **THREE** group members.
- Group members must be from the same lab.
- Groups must be finalised and recorded by your lab tutor by the end of week 8.
- Students will be randomly allocated to a group if they have not made a group by the end of week 8 and will be notified via email.
- If there are any issues with group members, it is recommended to report them immediately to your lab tutor in order to resolve them as soon as possible.

IMPORTANT

There are lots of tutorials and code freely available on the internet. If you do follow any tutorials or other sources, you should clearly provide a link in your documentation stating how much of your code is from the tutorial or other source. Failure to do so will result in marks being deducted from the Code section (see marking criteria on pages 4 and 5).

Please refer to Section 6 of the module handbook with regards to integrity and academic misconduct.

Coursework Weighting

50% of the module.

What to Submit

ONE PERSON from the group should submit a zipped folder containing all of the files and folders as specified in this brief.

IMPORTANT: To enable consistency and ease of marking the folder should be named **CST1500CW2** and the files should be neatly and sensibly arranged in folders.

You are required to record a video demo – name the video file sensibly e.g. CW2Demo.mp4.

The submission will be a zipped folder which contains:

The **CST1500CW2** folder containing:

- A Code folder – containing **ALL** of the code for each algorithm implemented.
- A Word/PDF document.
- The completed contribution spreadsheet.
- A video demonstration.

The following image is an example of the contents of the **CST1500CW2** folder and is similar to what your submission may look like:

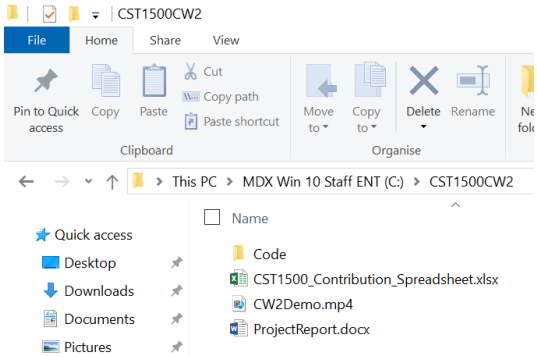

Fig. 1: Example folder contents.

Deadline

Friday 13th December 2024 18:00 (Week 12).

Where to submit

Submit your zipped folder to the dedicated submission link in the Coursework Two Submission folder on the course page.

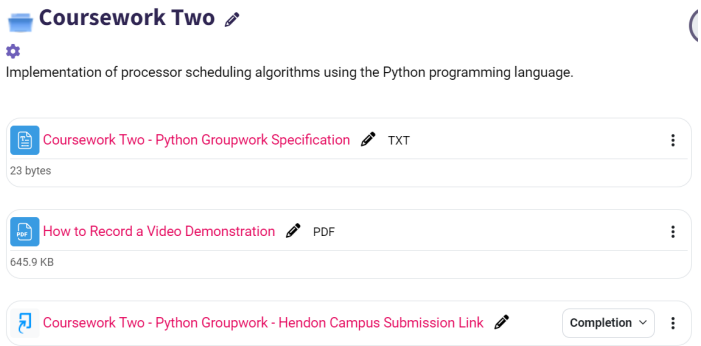

Fig. 2: Coursework Two submission link on the CST1500 course page.

Demonstration of Group Work

Your group work must be demonstrated in a video.

ALL MEMBERS OF THE GROUP SHOULD BE PRESENT IN THE VIDEO. This can be either on camera or by way of introducing themselves at the beginning of the video by saying their name and student ID clearly.

The demo will run your program and show the functionality.

You can present code, complex or clever code and discuss the program ONLY after you have presented the functionality and if you have sufficient time.

Critical reflection, group issues etc. should be included in the Word/PDF document, which will be marked after submission.

IMPORTANT

The video has a limit of 10 minutes – this will be an adequate amount of time to cover the functionality. Only the first 10 minutes will be watched by your lab tutor, anything after 10 minutes will be ignored. Therefore, be sure to cover the functionality adequately in the 10 minute video.

The file size of the video will increase exponentially, therefore consider the length of the video and final file size.

Each group member should contribute to the video demo and clearly state their name and student ID.

Failure to appear as part of the video demo will receive a zero for the demonstration contribution element of the contribution spreadsheet and therefore zero marks for the demonstration element (as shown in the marking section on page 6).

Details of how to record a video demonstration can be found on the course page in the Coursework Two section.

Therefore, you are strongly advised to contribute to the video demonstration.

Marking

This piece of coursework is worth **50%** of the overall mark for CST1500.

The **50%** is scaled to 100 marks and is distributed and applied using the software specific criteria shown on page 6.

To work out the final grade in terms of the 50% weighting:

Your final grade / 100 * 50

e.g. 80 / 100 * 50 = 40% - which you then add to your CW1 mark to calculate an overall (provisional) mark for the module.

IMPORTANT

This is a Group Project. However, marks will be awarded individually i.e. not one mark for the whole group. This is deemed to be fairer to students and mitigates a lot of issues regarding who contributed their 'fair share' to the project.

To implement this, a contribution spreadsheet will be used and **MUST** be submitted as a part of the final coursework submission. The contribution spreadsheet is available on the course page in the Coursework Two folder. You must download the spreadsheet, update the details to those of your group members and fill in the contribution columns. Once your marks have been awarded by the tutor, a final mark for each student will be calculated based on the contribution of each student.

The contribution distribution must add up to 1 and the 'Contribution Check' column must display 'Ok'. For example: Peter did all the work on the FCFS code so enters 1 in the 'Peter Contribution' cell of the FCFS row. Peter and Mary each did half of the work on the SJF code and so 0.5 is entered in the 'Peter Contribution' cell of the SJF row and 0.5 is also entered in the 'Mary Contribution' cell of the SJF row.

HINTS

- If all group members contribute equally to an element of the coursework enter =1/3 in each of the contribution cells. If it doesn't work and you get a date appear, try a different cell and copy/paste the result into the other relevant cells.
- If one member does not contribute at all, enter 0 in their contribution cell.
- You can divide the contribution as required, as long as the final result adds up to 1.

IMPORTANT

If there are any issues with group members, it is recommended to report them immediately to your lab tutor in order to resolve them as soon as possible.

The final contributions must be agreed upon by all members of the group before submission. If this cannot be resolved, enter =1/3 in each affected contribution cell. Then, make it clear in the report in a section entitled 'GROUP CONTRIBUTION CONFLICT' what the issues were and the lab tutor will try to resolve the contribution distribution with the group.

Contribution Spreadsheet Walkthrough

First you need to enter the names of the group members at the top of the sheet. Then you need to complete the pink fields for each task for each group member with a number that adds up to 1.0, or to 0.0 if the task was not attempted. If the numbers add up to 1.0 the contribution check field will display 'Ok'. If the numbers add up to 0.0, the contribution check field will display "Not implemented". Otherwise it will display "Error" and needs to be fixed.

Make sure that you enter the information carefully. If there are missing contribution values, we will assume that all group members made an equal contribution.

The contribution values for each task are multiplied by the mark awarded for each task to get the number of marks that each person contributed to the project. The contributed marks are used to calculate the individual marks from the group mark in the following way:

The variables are as follows:
• ic. Individual mark contribution
• gm. Group mark
• ac. Average mark contribution of the group members
• im. Final individual mark

If someone has made zero contribution, they will get zero marks for the project. Otherwise, their individual mark will be calculated from the group mark using the following formula:

im = gm + gm * (ic-ac)/ac

The result of this formula, im, will be limited to the maximum number of marks. For example:

Here the total is 70 (gm) from 100 – 70 (gm) is the maximum number of marks.
Student A has an im of 80.
80 exceeds 70 (gm), so Student A's Final Individual Mark is 70.

Student B has an im of 65. Therefore, Student B's Final Individual Mark is 65.

Most of the cells in the spreadsheet are locked to prevent the accidental introduction of errors.

There are separate sheets for groups of two and three people.

Marking Criteria

Item	1-4	5-8	9-12	13-16	Fail (17-20)
FCFS Code 25	High quality code exhibiting deep knowledge and understanding of Python and FCFS (comprehensive comments in code)	Code shows good structure, good understanding of Python and FCFS with good personal initiative (good comments in code)	Code shows some personal initiative and good understanding of basic Python and FCFS concepts (some comments in code)	Code works, mostly based on Python and FCFS examples with little personal contribution (a few comments in code)	Code does not work, and shows little understanding of Python and FCFS algorithm (no comments in code)
SJF Code 25	High quality code exhibiting deep knowledge and understanding of Python and SJF (comprehensive comments in code)	Code shows good structure, good understanding of Python and SJF with good personal initiative (good comments in code)	Code shows some personal initiative and good understanding of basic Python and SJF concepts (some comments in code)	Code works, mostly based on Python and SJF examples with little personal contribution (a few comments in code)	Code does not work, and shows little understanding of Python and SJF algorithm (no comments in code)
RR Code 25	High quality code exhibiting deep knowledge and understanding of Python and RR (comprehensive comments in code)	Code shows good structure, good understanding of Python and RR with good personal initiative (good comments in code)	Code shows some personal initiative and good understanding of basic Python and RR concepts (some comments in code)	Code works, mostly based on Python and RR examples with little personal contribution (a few comments in code)	Code does not work, and shows little understanding of Python and RR algorithm (no comments in code)
Documentation 15	Very well-presented documentation that covers the whole project	Satisfactory presentation. Documentation covers most of the project	Basic but correct documentation which covers some elements of the project	Poor documentation missing lots of elements of the project	Missing or minimal documentation
Demonstration 10	Very clear and organised demonstration of code (practise evident)	Clear and organised demonstration of code	Demonstration is clear but not very organised	Basic explanation of code	No clear explanation, lack of understanding. No lab demo or video demo

Detailed Specification

Your group is required to write a series of programs using the Python programming language. The programs should demonstrate up to three CPU scheduling algorithms from the following:

First Come First Served (FCFS).
Shortest Job First (SJF).
Round Robin (RR).

Your group can pick one, two or three of the algorithms in any combination.

We will be looking for well structured code which uses Python programming conventions, sensible naming conventions and contains comprehensive code comments.

We will also be looking for correct output from the code, or in the case of partially working code, the expected output should be displayed in your report in the form of a table.

Each algorithm that you write should correctly implement the functionality according to its specification. You can find brief details of each algorithm in the lecture video Lecture 13 CPU Scheduling on the course page. **Further investigation is recommended**.

CPU Algorithm ONE: First Come First Served (FCFS)

Your program should output:

Process Number
Burst Time
Waiting Time
Turn Around Time
Average Waiting Time
Average Turn Around Time

AT A MINIMUM your program should provide output for the following input:

Process Number	Burst Time
1	5
2	8
3	12

User input is recommended to improve the potential for higher marks.

CPU Algorithm TWO: (Non-Pre-emptive) Shortest Job First (SJF)

Your program should output:

Process Number
Burst Time
Waiting Time
Turn Around Time
Average Waiting Time
Average Turn Around Time

AT A MINIMUM your program should provide output for the following input:

Process Number	Burst Time
1	5
2	3
3	7

User input is recommended to improve the potential for higher marks.

CPU Algorithm THREE: Round Robin (RR)

Your program should output:

Process Number
Arrival Time
Burst Time
Completion Time
Turn Around Time
Waiting Time
Average Waiting Time
Average Turn Around Time

AT A MINIMUM your program should provide output for the following input:

Process Number	Burst Time
1	5
2	3
3	7
4	2

The Arrival Time should be considered as 0 for all processes.
The Time Quantum should be considered as 2 for all processes.
*If user input is implemented, Arrival Time and Time Quantum can be user defined.

User input is recommended to improve the potential for higher marks.

Documentation

You must submit a Word or PDF document that should contain the following sections:

- Introduction – group members names and student IDs, contribution of each group member i.e. tasks that each member undertook.
- CPU Scheduling Algorithms – which of the three algorithms your group implemented with a brief description of each one (**ONLY** the algorithms that your group implemented).
- Final Output – screenshots of the code and the output of the programs with brief labels. If the code is incomplete or does not produce correct output, the output screenshots only can be replaced with the expected output in the form of a table (Code screenshots should still be included).
- Conclusion – a summary of the project, what went well, what you would have done differently, any problems your group faced etc.

Make sure that your document is well-presented i.e. a cover page, contents page, sections, good spelling and grammar, labels on images/tables etc. This document is likely to be around 1000 - 1500 words. There is no word limit, however if you find that the document is substantially over 1500 words, you may be doing too much.